ESG 633 Programming for Geoinformatics
Lab 6: Buffer and Overlay Analysis

Note: Refer https://github.com/rlessgeoai-design/MTech26/tree/main/lab%206

1. Generate random points using geopandas and create a buffer of 100m, 500m and 1km around the points. Plot the results using Matplotlib.
2. Generate a circular polygon and a rectangular polygon with overlapping shapes using Geopandas. Overlay these polygons using geopandas overlay function and plot the results.
3. **Explore the given data** (print results like count of forests from LULC data, Primary road names from road data etc. It can be of your choice) and **print the area of Nedumangad**
4. Find the buildings that are within **200m distance of rivers and streams** and print the name of the buildings (10 names). Plot the results.
5. Find the hospitals that are within **750m distance of primary and trunk roads** and print the name of the hospitals (10 names). Plot the results.
6. Find the **bridges that are on top of rivers**. Print the total number of bridges and names of the bridge.
7. Find the **primary and tertiary roads that are inside the forest**. Print the names of the roads that are inside the forest.